NOMAD TORONTO — SOUND SYSTEM SPECIFICATION

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1. SYSTEM OVERVIEW

The Nomad Toronto sound system is a VOID Acoustics installation built around a single amplifier rack housing six Bias-platform DSP amplifiers, a Drawmer SP2120 speaker protection processor, and a Tripp Lite power distribution unit. The system serves four distinct zones: main dancefloor (bi-amped mains plus sub arrays), outboard flank subs, DJ stage monitoring, and mid-room fill coverage. All amplifiers are networked for remote management via Powersoft Armonía Pro Audio Suite.

The loudspeaker complement is entirely VOID Acoustics, drawing from the Air Series (Air Motion V2 Red, Air Vantage), the Stasys Series (Stasys Xair), the Venu Series (Venu 215 V2), and the Cyclone Series (Cyclone fills). The system is designed for high-output electronic music playback in a nightclub environment with emphasis on sub-bass impact and consistent dancefloor coverage.

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2. LOUDSPEAKER INVENTORY

2.1 Mains — VOID Acoustics Air Motion V2 Red (x2, Left/Right)

The Air Motion V2 is the flagship of VOID's Air Series — a three-way, bi-amped sculpted loudspeaker array with isometric conical horn loading on all three transducers. The "Red" designation indicates the custom colour variant. These are mirror-image Left and Right cabinets. Because they are bi-amped, each cabinet requires two amplifier channels and two NL4 speaker cables: one for LF and one for HMF.

Specification (per cabinet):

Frequency Response: 140 Hz – 20 kHz ±3 dB. Efficiency: LF section 106 dB 1W/1m, HMF section 108 dB 1W/1m. Nominal Impedance: LF 8 Ω, HMF 8 Ω. Power Handling: LF 500 W AES, HMF 250 W AES (750 W AES total per cabinet). Maximum Output: 132 dB continuous, 138 dB peak. Driver Configuration: 1 × 12″ LF transducer, 1 × 8″ MF transducer, 1 × 1.5″ HF compression driver (MF and HF crossed over passively within the HMF section). Dispersion: 60° horizontal × 50° vertical. Connectors: 2 × 4-pole Neutrik speakON NL4 (NL4 #1 = LF section, NL4 #2 = HMF section). Weight: 35.4 kg (78 lbs). Enclosure: Fibreglass Kevlar composite. Finish: Custom Red.

Amplifier Assignment: LF section driven by Bias V9 #1 (CH1 = Left, CH2 = Right) via NL4 #1. HMF section driven by Bias Q2 #1 (CH1 = Left, CH2 = Right) via NL4 #2. Both connections must be present and correct for each cabinet; swapping LF and HMF cables puts the wrong frequency band on the wrong drivers.

2.2 Subwoofers — VOID Acoustics Stasys Xair (x4)

The Stasys Xair is the Air Series edition of VOID's acclaimed Stasys XV2. It is a double 18″ horn-loaded subwoofer with exceptional low-frequency output and negligible power compression. Four units are deployed as two stacked pairs (Left stack: Xair L-1 + L-2, Right stack: Xair R-1 + R-2), each pair wired in parallel on a single amplifier channel, presenting a 2Ω load.

Specification (per cabinet):

Frequency Response: 30 Hz – 180 Hz ±3 dB. Efficiency: 103.8 dB 1W/1m (half-space). Crossover Points: 38 Hz high-pass 24 dB/oct, 100–145 Hz low-pass 24 dB/oct (active, set in amplifier DSP). Nominal Impedance: 4 Ω (single cabinet), 2 Ω (two cabinets paralleled). Power Handling: 3,200 W AES per cabinet. Maximum Output: 139 dB continuous, 145 dB peak (per cabinet). Driver Configuration: 2 × 18″ LF horn-loaded transducers (neodymium magnet, 4″ dual voice coil, cast aluminium frame). Connectors: 2 × 4-pole Neutrik speakON NL4 (one input, one link-out for parallel connection to second cabinet). Weight: 130 kg (286.6 lbs). Enclosure: 18 mm birch plywood. Finish: Textured TourCoat polyurea with Incubus-style aluminium bracing.

Amplifier Assignment: Left stack (L-1 + L-2 paralleled) on Bias V3 #1 CH1, 2Ω parallel load. Right stack (R-1 + R-2 paralleled) on Bias V3 #2 CH1, 2Ω parallel load. 2Ω operating mode must be enabled on both V3 amplifiers for these channels.

2.3 Outboard Flank Subs — VOID Acoustics Venu 215 V2 (x2, Left/Right)

The Venu 215 V2 is a reflex-loaded dual 15″ low-frequency enclosure from the Venu Series, designed for installation and touring applications. In this system, the two Venu 215 V2 cabinets are deployed as outboard flank subs — positioned further from the main sub stacks to extend bass coverage across the room width.

Specification (per cabinet):

Frequency Response: 38 Hz – 160 Hz ±3 dB. Efficiency: 101 dB 1W/1m (half-space). Crossover Points: 80–160 Hz low-pass active (set in amplifier DSP). Nominal Impedance: 4 Ω. Power Handling: 1,000 W AES. Maximum Output: 134 dB continuous, 140 dB peak. Driver Configuration: 2 × 15″ LF transducers (high excursion, 4″ voice coil). Connectors: 1 × Phoenix terminal block with link-out, plus 1 × speakON NL4 with link-out (recessed panel). Weight: 62.5 kg (137.8 lbs). Enclosure: 15 mm birch plywood. Finish: Textured polyurethane.

Critical Connector Note: The Venu 215 V2 has both Phoenix and speakON connectors on its recessed rear panel. The V3 amplifier outputs are NL4 speakON, so direct NL4-to-NL4 connection is possible if the speakON port is used. If the Phoenix port is used instead, an adapter cable is required.

Amplifier Assignment: Left flank on Bias V3 #1 CH2. Right flank on Bias V3 #2 CH2.

2.4 DJ Stage Monitors — VOID Acoustics Air Vantage (x2, Left/Right)

The Air Vantage is a dual-purpose loudspeaker from the Air Series, serving as either a front-of-house mid-top or a DJ monitor. It features a coaxial 12″ / 1.5″ driver arrangement with true point source radiation and a passive internal crossover optimised for minimal listener fatigue during extended high-SPL DJ sets.

Specification (per cabinet):

Frequency Response: 140 Hz – 20 kHz ±3 dB. Efficiency: 100 dB 1W/1m. Crossover Points: 1.6 kHz passive (internal). Nominal Impedance: 8 Ω. Power Handling: 500 W AES. Maximum Output: 126 dB continuous, 132 dB peak. Driver Configuration: 1 × 12″ LF, 1 × 1.5″ HF coaxial neodymium compression driver. Dispersion: 70° horizontal × 40° vertical. Connectors: 1 × 4-pole Neutrik speakON NL4. Weight: 23.5 kg (51.8 lbs). Enclosure: Moulded fibreglass composite with smooth cellulose finish.

Amplifier Assignment: Left monitor on Bias Q2 #2 CH1. Right monitor on Bias Q2 #2 CH2. The Q2 amplifier outputs via Phoenix terminal block — an adapter cable from Phoenix to NL4 is required for connection to the Air Vantage's speakON input.

2.5 Mid-Room Fills — VOID Acoustics Cyclone 2 (x2, Left/Right)

The system uses two Cyclone series loudspeakers as mid-room fills to provide coverage in the intermediate zone between the DJ booth and the main dancefloor. Based on the Cyclone series naming convention and their deployment as full-range fills, these are most likely the Cyclone 10 (10″ coaxial, full-range) or a similar Cyclone variant. The specifications below are based on the Cyclone 10 as the closest match; confirm the exact model on-site.

Specification (per cabinet, Cyclone 10 assumed):

Frequency Response: 52 Hz – 22 kHz ±3 dB. Efficiency: 97 dB 1W/1m. Crossover Points: 2.1

kHz passive. Nominal Impedance: 8 Ω. Power Handling: 350 W AES. Maximum Output: 122 dB continuous, 128 dB peak. Driver Configuration: 1 × 10″ LF, 1 × 1″ HF compression driver (coaxial). Dispersion: 90° horizontal × 60° vertical. Connectors: Phoenix terminal block with link-out. IP Rating: IP-55 (BS EN 60529:1992 +A2:2013). Weight: 14.5 kg (32 lbs). Enclosure: Fibreglass with smooth cellulose finish. Mounting: Type 75 plate.

Amplifier Assignment: Left fill on Bias Q2 #3 CH1. Right fill on Bias Q2 #3 CH2. The Q2 outputs and Cyclone inputs are both Phoenix terminal blocks — direct connection is possible without adapters.

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3. AMPLIFIER RACK — EQUIPMENT INVENTORY

The amplifier rack is a standard 19″ open-frame rack mounted on a wooden riser, providing cable storage space beneath the rack. Equipment is installed from Position 2 (bottom) to Position 10 (top). Position 1 is empty or occupied by rear rack rail.

3.1 Rack Layout (Top to Bottom, Rear View)

Position 10 — Tripp Lite PDU, 1U. Power distribution for control equipment only (SP2120, control PC). Amplifiers are NOT powered from this PDU.

Position 9 — Empty / TBD. No network switch was identified in the photos. Teal Cat5e/Cat6 cables are visible running from amplifier AESOP/Ethernet ports; their destination (external switch or direct PC connection) is to be confirmed on-site.

Position 8 — VOID Acoustics Bias Q2 #3, 1U. Mid-room fill amplifier driving Cyclone 2 Left (CH1) and Cyclone 2 Right (CH2).

Position 7 — VOID Acoustics Bias V3 #2, 2U. Right sub array amplifier driving Xair R-1 + R-2 paralleled at 2Ω (CH1) and VENU 215 V2 Right (CH2).

Position 6 — VOID Acoustics Bias Q2 #2, 1U. DJ monitor amplifier driving Air Vantage Left (CH1) and Air Vantage Right (CH2).

Position 5 — VOID Acoustics Bias V9 #1, 2U. Mains low-frequency amplifier driving Air Motion V2 Red Left LF section (CH1) and Air Motion V2 Red Right LF section (CH2). This is the highest-current amplifier in the rack.

Position 4 — VOID Acoustics Bias Q2 #1, 1U. Mains high-mid frequency amplifier driving Air Motion V2 Red Left HMF section (CH1) and Air Motion V2 Red Right HMF section (CH2).

Position 3 — VOID Acoustics Bias V3 #1, 2U. Left sub array amplifier driving Xair L-1 + L-2 paralleled at 2Ω (CH1) and VENU 215 V2 Left (CH2).

Position 2 — Drawmer SP2120, 1U. Speaker protection processor, positioned directly before the amplifiers in the signal chain.

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4. AMPLIFIER SPECIFICATIONS

4.1 VOID Acoustics Bias V3 (Positions 3 & 7) — 2-Channel DSP Amplifier, 2U

The Bias V3 is a 2-channel DSP power amplifier with Powersoft PFC (Power Factor Correction) power supply, built-in DSP processing, and Armonía Pro Audio Suite remote management. It provides line-level outputs (pre-DSP) for signal distribution to downstream amplifiers.

Rear Panel I/O (confirmed from manufacturer manual V1.0):

Mains Power: IEC C20 inlet, universal PFC supply 100–240 VAC ±10%, maximum current 8–16A (user-settable via rear panel selector). Analog Inputs: 2 × Neutrik XLR-F, balanced (pin 1 GND, pin 2 Hot, pin 3 Cold). Input 2 switchable to AES3 digital via rear toggle. Line Outputs: 2 × Neutrik XLR-M, pre-DSP replica of analog input signal. Speaker Outputs: 2 × Neutrik NL4MD speakON (wiring: pins 1+/2+ bridged to positive, pins 1-/2- bridged to negative). Network: 2 × RJ45 AESOP ports on rear (primary, carries data + AES3 streams), additional secondary ports on front (data only). External Voltage: Phoenix 2-pin Vext connector (12VDC, 1A max, AESOP version). Link Button: Bridges CH1 and CH2 inputs for bridged mono operation.

Key Operational Notes: The V3's line outputs are a critical feature in this system — they can daisy-chain signal from the V3 to downstream amps (V9, Q2) without requiring an external distribution amplifier. The 2Ω parallel sub loads on CH1 require the amplifier's low-impedance operating mode to be enabled via either the rear DIP switches or Armonía software.

4.2 VOID Acoustics Bias V9 (Position 5) — 2-Channel DSP Amplifier, 2U

The Bias V9 is the high-power sibling of the V3, sharing the same DSP platform and Armonía integration but with a significantly higher current capacity and an industrial mains connector instead of IEC.

Rear Panel I/O (confirmed from manufacturer manual V1.0):

Mains Power: AMP CPC 45A industrial connector (NOT IEC C20), universal PFC supply 100–240 VAC ±10%, maximum current 15–32A (user-settable). This is a permanently wired connection — there is no plug-in IEC cord. Cable specification is LAPP OLFLEX 191 3G6 or equivalent SJT 3×AWG10. Analog Inputs: 2 × Neutrik XLR/TRS hybrid combo connectors (accepts either XLR or 1/4″ TRS balanced). Input 2 switchable to AES3. Line Outputs: NONE — the V9 does not have line-level outputs unlike the V3. This is a critical difference that affects signal distribution architecture. Speaker Outputs: 2 × Neutrik NL4MD speakON (same wiring as V3). Network: 2 × RJ45 AESOP ports (rear, primary). External Voltage: Phoenix 2-pin Vext (12VDC, 1A). Link Button: Available for bridged operation.

Key Operational Notes: The V9's CPC 45A connector is the large industrial multi-pin connector visible on the left side of the rack in the photos — it is distinctly different from the IEC C20 connectors on the other amplifiers. The absence of line outputs means the signal chain cannot daisy-chain through the V9 to reach downstream amplifiers.

4.3 VOID Acoustics Bias Q2 (Positions 4, 6, 8) — 4-Channel DSP Amplifier, 1U

The Bias Q2 is a compact 4-channel DSP amplifier from the same Bias/Powersoft platform. In this system, only 2 of the 4 available channels are used per unit (CH1 and CH2), as each Q2 is assigned a stereo pair of loudspeakers. The Q2 uses Phoenix terminal block connectors for both input and output — NOT XLR or speakON — which is the most important distinction from the V3/V9 amplifiers.

Rear Panel I/O (confirmed from manufacturer manual V1.0):

Mains Power: IEC C20 inlet, universal PFC supply 100–240 VAC ±10%. Analog Inputs: Phoenix MC 1,5/6-ST-3,81 terminal block connector (6-pin, balanced analog). This is NOT an XLR connector. Adapter cables (XLR-to-Phoenix or bare-wire-to-Phoenix) are required to interface with standard XLR source equipment. Speaker Outputs: Phoenix PC 5/8-STF1-7,62 terminal block connector (8-pin for 4 channels). This is NOT a speakON connector. Adapter cables are required for loudspeakers with NL4 speakON inputs (Air Motion, Air Vantage). Loudspeakers with native Phoenix inputs (Cyclone series, Venu series) can connect directly. DIP Switches (per channel): Gain selection (26/29/32/35 dB), Lo-Z/Hi-Z mode, High-Pass Filter (35 Hz / 70 Hz), 2Ω mode, Energy Save, Breaker Save. Remote On/Off: Phoenix MC 1,5/4-ST-3,81 connector (4-pin), responds to 5–24 VDC differential. Do not exceed 28 VDC. Remote Level: Per-channel potentiometer connector for external volume control. GPO/Alarm: Phoenix MC 1,5/6-ST-3,81 per channel pair (NO/NC relay contacts) for fault reporting. Network: RJ45 Ethernet (for Armonía control). Dante: RJ45 (for digital audio, DSP+D versions only).

Key Operational Notes: The Phoenix terminal block connectors are the green blocks visible in the rear photos of the rack. Every connection to and from a Q2 amplifier requires either a Phoenix-terminated cable or an adapter. The DIP switches on the rear panel are the primary means of configuring gain and operating mode when Armonía is not connected.

4.4 Drawmer SP2120 (Position 2) — 2-Channel Speaker Protector, 1U

The Drawmer SP2120 is a tamper-proof speaker protection processor designed to prevent excessive sound pressure levels from damaging loudspeakers and amplifiers. It uses dynamic processing (limiting) rather than hard switching or power disconnection, providing transparent level control. A key-lock security mechanism on the front panel ensures only authorised personnel (venue manager, maintenance engineer) can adjust the protection thresholds.

Rear Panel I/O (confirmed from manufacturer manual):

Analog Inputs: 2 × XLR-F, electronically balanced (pin 1 GND, pin 2 Hot, pin 3 Cold), input impedance 20 kΩ, maximum input level +21 dBu. Analog Outputs: 2 × XLR-M, electronically balanced, output impedance 100 Ω, maximum output level variable from infinity (mute) to +14 dBu. Optional output isolation transformers can be fitted. Mains Power: IEC inlet, 115V/230V manually switchable, fused T50mA at 230V / T100mA at 115V, power consumption 9VA. Tamper-Proof Bracket: Optional rear bracket (serial 0845+) that traps the XLR connectors in place, preventing disconnection without bracket removal. This is recommended for compliance with noise abatement regulations.

Front Panel Controls: Key-lock security switch, VU meter calibrate preset (sets the signal level at which the bargraph display reaches full scale), maximum output level preset (sets the absolute ceiling the SP2120 will allow through), 16-segment Left/Right LED bargraphs, protection activity meter.

Key Operational Notes: The SP2120 has only 2 outputs (Left and Right). The system has 6 amplifiers requiring signal. This creates a signal distribution challenge — the SP2120 cannot directly feed all amplifiers. The current architecture likely relies on the V3 amplifiers' line outputs to daisy-chain signal downstream, or uses an external distribution amplifier or passive splitter not visible in the rack.

4.5 Tripp Lite PDU (Position 10) — Power Distribution Unit, 1U

Standard rack-mount PDU with NEMA receptacles. Powers control-class equipment only: the Drawmer SP2120 and the Armonía control PC. All six amplifiers are powered from dedicated panel circuits with individual breakers — they are NOT on this PDU.

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5. SIGNAL FLOW ARCHITECTURE

5.1 Source to Protection

The DJ mixer main stereo output (Left/Right) feeds the Drawmer SP2120 via balanced XLR cables. The SP2120 applies tamper-proof level limiting and outputs a protected stereo signal.

5.2 Protection to Amplifiers

The SP2120's two outputs must be distributed to all amplifiers that require the main dancefloor signal. The confirmed and assumed routing is as follows:

SP2120 Output L → Bias V3 #1 Input 1 (XLR). Bias V3 #1 Line Out CH1 (XLR-M, pre-DSP) → Bias V9 #1 Input 1 (Combo) for mains LF. Bias V3 #1 Line Out CH2 (XLR-M, pre-DSP) → Bias Q2 #1 Input CH1 (Phoenix, via adapter) for mains HMF.

SP2120 Output R → Bias V3 #2 Input 1 (XLR). Bias V3 #2 Line Out carries the right-channel signal similarly.

DJ monitor and fill feeds (Bias Q2 #2 and Q2 #3) may receive signal from a separate mixer auxiliary bus, a passive mult/split, or from the V3 line outputs via additional Y-splits. The exact routing for these zones should be confirmed on-site, as it determines whether the SP2120 protection extends to monitors and fills or only to subs and mains.

5.3 Amplifiers to Loudspeakers

Sub Arrays (V3 #1 and V3 #2 speaker outputs): CH1 NL4 → Xair sub stacks (parallel pairs, 2Ω). CH2 NL4 → VENU 215 V2 flank subs (NL4 or Phoenix connection depending on which cabinet connector is used).

Mains LF (V9 #1 speaker outputs): CH1 NL4 → Air Motion V2 Red Left, NL4 #1 port (LF section). CH2 NL4 → Air Motion V2 Red Right, NL4 #1 port (LF section).

Mains HMF (Q2 #1 speaker outputs): CH1 Phoenix → Air Motion V2 Red Left, NL4 #2 port (HMF section), via Phoenix-to-NL4 adapter cable. CH2 Phoenix → Air Motion V2 Red Right, NL4 #2 port (HMF section), via Phoenix-to-NL4 adapter cable.

DJ Monitors (Q2 #2 speaker outputs): CH1 Phoenix → Air Vantage Left, NL4 port, via Phoenix-to-NL4 adapter. CH2 Phoenix → Air Vantage Right, NL4 port, via Phoenix-to-NL4 adapter.

Mid-Room Fills (Q2 #3 speaker outputs): CH1 Phoenix → Cyclone 2 Left, Phoenix input (direct connection). CH2 Phoenix → Cyclone 2 Right, Phoenix input (direct connection).

5.4 Network / Control

All six amplifiers connect via RJ45 (Cat5e/Cat6, T568B straight-through) to the venue's network infrastructure for Armonía Pro Audio Suite remote management. The V3 and V9 amplifiers use AESOP-protocol ports (which carry both control data and AES3 digital audio streams). The Q2 amplifiers use standard Ethernet ports for Armonía control, plus a separate Dante port on DSP+D versions for digital audio transport. Armonía's AESOP protocol warns against daisy-chaining in production environments; star topology (each amp home-run to a switch) is recommended.

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6. POWER DISTRIBUTION

6.1 Amplifier Power (Dedicated Panel Circuits)

Each amplifier is fed from its own dedicated breaker on the venue's electrical panel. This is the correct approach for high-current amplifiers to avoid shared-circuit voltage sag, breaker trips, and ground loops.

Bias V3 #1 (Pos 3): IEC C20, 100–240 VAC, max 16A. Breaker: 20A C-curve or D-curve recommended. Bias Q2 #1 (Pos 4): IEC C20, 100–240 VAC, max 16A. Breaker: 20A C-curve or D-curve. Bias V9 #1 (Pos 5): CPC 45A, 100–240 VAC, max 32A. Breaker: 40A C-curve or D-curve. This is the heaviest draw in the rack. Bias Q2 #2 (Pos 6): IEC C20, 100–240 VAC, max 16A. Breaker: 20A. Bias V3 #2 (Pos 7): IEC C20, 100–240 VAC, max 16A. Breaker: 20A. Bias Q2 #3 (Pos 8): IEC C20, 100–240 VAC, max 16A. Breaker: 20A.

6.2 Control Power (Tripp Lite PDU)

The PDU is fed from a separate control circuit and powers only low-draw equipment: the Drawmer SP2120 (9VA), the control PC, and any other low-current accessories. Total draw on this circuit is well under 5A.

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7. LOAD SUMMARY TABLE

Zone	Cabinet	Qty	Amplifier	Channel	Load (Ω)	Power (W AES)	Connector (Amp→Cab)
Left Sub Stack	Stasys Xair L-1 + L-2	2	Bias V3 #1	CH1	2Ω (parallel)	3,200 × 2	NL4 → NL4 (link-out)
Left Flank Sub	Venu 215 V2 Left	1	Bias V3 #1	CH2	4Ω	1,000	NL4 → NL4 or Phoenix
Mains LF Left	Air Motion V2 Red	1	Bias V9 #1	CH1	8Ω	500	NL4 → NL4 #1

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Mains LF Right	Air Motion V2 Red R	1	Bias V9 #1	CH2	8Ω	500	NL4 → NL4 #1
Mains HMF Left	Air Motion V2 Red L	1	Bias Q2 #1	CH1	8Ω	250	Phoenix → NL4 #2 (adapter)
Mains HMF Right	Air Motion V2 Red R	1	Bias Q2 #1	CH2	8Ω	250	Phoenix → NL4 #2 (adapter)
DJ Monitor Left	Air Vantage Left	1	Bias Q2 #2	CH1	8Ω	500	Phoenix → NL4 (adapter)
DJ Monitor Right	Air Vantage Right	1	Bias Q2 #2	CH2	8Ω	500	Phoenix → NL4 (adapter)
Right Sub Stack	Stasys Xair R-1 + R-2	2	Bias V3 #2	CH1	2Ω (parallel)	3,200 × 2	NL4 → NL4 (link-out)
Right Flank Sub	Venu 215 V2 Right	1	Bias V3 #2	CH2	4Ω	1,000	NL4 → NL4 or Phoenix
Fill Left	Cyclone 2 Left	1	Bias Q2 #3	CH1	8Ω	350	Phoenix → Phoenix (direct)
Fill Right	Cyclone 2 Right	1	Bias Q2 #3	CH2	8Ω	350	Phoenix → Phoenix (direct)

Total Loudspeaker Count: 14 cabinets (4 Xair, 2 Venu 215 V2, 2 Air Motion V2 Red, 2 Air Vantage, 2 Cyclone 2). **Total Amplifier Channels in Use:** 12 of 18 available (6 amps × 2 or 4 channels each; Q2s have 4 channels but only 2 are used per unit, leaving 6 spare Q2 channels).

8. CONNECTOR & ADAPTER CABLE REQUIREMENTS

The mixture of Neutrik XLR, Neutrik speakON NL4, and Phoenix terminal block connectors across the amplifiers and loudspeakers requires several adapter cables. A complete system requires the following custom or adapter cables beyond standard XLR and NL4 stock:

XLR-M to Phoenix MC 1,5/6-ST-3,81 (signal adapters for Q2 inputs): Required wherever a standard XLR source feeds a Q2 amplifier's Phoenix input. Quantity depends on how many Q2 inputs receive analog signal (up to 6 cables for Q2 #1, #2, and #3 at 2 channels each).

Phoenix PC 5/4-STF1-7,62 to Neutrik NL4 (speaker adapters for Q2 outputs): Required for Q2 outputs feeding loudspeakers with NL4 speakON inputs (Air Motion HMF, Air Vantage). Four required: 2 for Q2 #1 → Air Motion V2 Red HMF, 2 for Q2 #2 → Air Vantage.

Phoenix PC 5/4-STF1-7,62 to Phoenix (speaker cables for Q2 outputs to Cyclone): The Cyclone series uses native Phoenix connectors, so Q2 #3 can connect directly using standard Phoenix-to-Phoenix speaker cable. Two required.

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9. DSP & NETWORK CONFIGURATION NOTES

All Bias amplifiers should be loaded with VOID Acoustics factory DSP presets for each connected loudspeaker model. These presets include crossover frequencies, driver protection limiters, EQ curves, and delay alignment. Presets are loaded and managed through Armonía Pro Audio Suite.

Critical DSP settings to verify:

V3 #1 CH1 and V3 #2 CH1: 2Ω operating mode enabled (for parallel Xair loads). Crossover set to sub-bass band per Xair preset (typically 30–100 Hz bandpass). V3 #1 CH2 and V3 #2 CH2: Crossover set per Venu 215 V2 preset (typically 38–160 Hz bandpass). V9 #1 CH1 and CH2: Crossover set per Air Motion V2 LF preset (bi-amp low band, typically 140 Hz low-pass). Q2 #1 CH1 and CH2: Crossover set per Air Motion V2 HMF preset (bi-amp high band, typically 140 Hz high-pass). Q2 #2 CH1 and CH2: Full-range Air Vantage preset (140 Hz – 20 kHz, high-passed at the bottom for use with external subs). Q2 #3 CH1 and CH2: Fill preset for Cyclone (full-range or high-passed depending on room requirements).

Network Configuration: All amplifiers default to DHCP/AutoIP. For production stability, fixed IP addressing is recommended. Armonía host PC and all amplifiers must be on the same subnet. If Dante is used on the Q2 DSP+D units, a separate VLAN or physically separate network is recommended to keep Dante audio traffic isolated from Armonía control traffic.

10. DOCUMENT REFERENCES

VOID Acoustics Bias V3/V9 User Guide V1.0 — d181c7pevbxova.cloudfront.net VOID Acoustics Bias Q2/Q1/D1 User Guide V1.0 — d181c7pevbxova.cloudfront.net VOID Acoustics Air Motion User Guide V2.0 (UG10519-2.0) VOID Acoustics Stasys Xair Specification Sheet V1.1 VOID Acoustics Venu 215 V2 Specification Sheet V1.0 VOID Acoustics Air Vantage Specification Sheet V1.0 VOID Acoustics Cyclone Bass / Cyclone 10 Specification Sheet Drawmer SP2120 Operator's Manual — drawmer.com Powersoft Armonía Pro Audio Suite documentation — voidacoustics.com